WORKED SOLUTION

STEP I 1999, Question 8

Pure • Harmonic Series Bounds

STEP I 1999, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q8

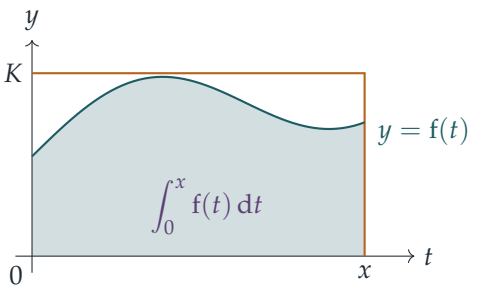
The key idea

An integral is an area, so trapping a curve inside a rectangle traps its area between 0 and the rectangle's area. The suggested integrand splits by partial fractions into something whose integral is exactly $\ln(\frac{n}{n-1}) - \frac{1}{n}$, and the bounds on the integrand give the inequality. After that, everything telescopes: adding up over n turns a chain of logarithms into $\ln N$ and a chain of differences of reciprocals into almost nothing.

Solution

Bounding an integral

Draw the graph of $y = \mathrm{f}(t)$ for $0 \le t \le x$. Because $0 \le \mathrm{f}(t) \le K$, the whole curve lies inside the rectangle with corners $(0, 0)$ and (x, K), on or above the t-axis.



The integral is the shaded area under the curve. That area can't be negative, since the curve never dips below the axis, and it can't exceed the area Kx of the rectangle, since the region sits inside it. So

$$0 \le \int_0^x \mathrm{f}(t)\,\mathrm{d}t \le Kx.$$

The inequality for $\ln\left(\frac{n}{n-1}\right)$

Take $n > 1$. For $0 \le t \le 1$ we have $n - t \ge n - 1 > 0$, so the integrand is defined. Splitting it into partial fractions,

$$\frac{t}{n(n-t)} = \frac{1}{n} \cdot \frac{n-(n-t)}{n-t} = \frac{1}{n-t} - \frac{1}{n},$$

so

$$\int_0^1 \frac{t}{n(n-t)}\,\mathrm{d}t = \Big[-\ln(n-t) - \frac{t}{n}\Big]_0^1 = \ln n - \ln(n-1) - \frac{1}{n} = \ln\Big(\frac{n}{n-1}\Big) - \frac{1}{n}.$$

Now we bound the integrand on $[0,1]$. The numerator t is non-negative and the denominator $n(n-t)$ is positive, so the integrand is at least 0. As t increases from 0 to 1 the numerator grows and the positive denominator shrinks, so the integrand is increasing and its largest value is at $t = 1$:

$$0 \le \frac{t}{n(n-t)} \le \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}.$$

Using the first result with $x = 1$ and $K = \frac{1}{n-1} - \frac{1}{n}$,

$$0 \le \ln\Big(\frac{n}{n-1}\Big) - \frac{1}{n} \le \frac{1}{n-1} - \frac{1}{n},$$

as required.

Summing from $n = 2$ to N

Add the inequality for $n = 2, 3, \ldots, N$ (where $N \ge 2$). The logarithms telescope, because

$$\sum_{n=2}^{N} \ln\Big(\frac{n}{n-1}\Big) = \ln\Big(\frac{2}{1} \cdot \frac{3}{2} \cdots \frac{N}{N-1}\Big) = \ln N,$$

and so do the right-hand sides:

$$\sum_{n=2}^{N}\Big(\frac{1}{n-1} - \frac{1}{n}\Big) = \Big(1 - \frac{1}{2}\Big) + \Big(\frac{1}{2} - \frac{1}{3}\Big) + \cdots + \Big(\frac{1}{N-1} - \frac{1}{N}\Big) = 1 - \frac{1}{N}.$$

Therefore

$$0 \le \ln N - \sum_{n=2}^{N} \frac{1}{n} \le 1 - \frac{1}{N} \le 1.$$

(For $N = 1$ the sum is empty and $\ln 1 = 0$, so the statement holds then too.)

Ask yourself Which half of this double inequality tells us the harmonic series diverges, and which half gives an upper bound?

The left half says the sum is *at most* $\ln N$; the right half says it is *at least* $\ln N - 1$. For divergence we need a lower bound, so we use the right half.

Divergence

From $\ln N - \sum_{n=2}^{N} \frac{1}{n} \leq 1$ we get $\sum_{n=2}^{N} \frac{1}{n} \geq \ln N - 1$, and adding the $n = 1$ term,

$$\sum_{n=1}^{N} \frac{1}{n} \geq \ln N.$$

Since $\ln N \to \infty$ as $N \to \infty$, the sum also tends to infinity.

The bound for $N < 10^{30}$

Now we use the left half: $\sum_{n=2}^{N} \frac{1}{n} \leq \ln N$, so

$$\sum_{n=1}^{N} \frac{1}{n} \leq 1 + \ln N.$$

Since $2^{10} = 1024 > 1000 = 10^3$, raising to the tenth power gives $10^{30} < 2^{100}$. So if $N < 10^{30}$ then $N < 2^{100}$ and, as ln is increasing,

$$\ln N < 100 \ln 2 < 100,$$

using $\ln 2 < 1$, which holds because $2 < \mathrm{e}$. Hence

$$\sum_{n=1}^{N} \frac{1}{n} \leq 1 + \ln N < 101.$$

Checks

- Simpson's rule on $\int_0^1 \frac{t}{n(n-t)}\,\mathrm{d}t$ agrees with $\ln\left(\frac{n}{n-1}\right) - \frac{1}{n}$ for many values of $n > 1$. For example, with $n = 2$ both give $\ln 2 - \frac{1}{2} \approx 0.193$, which lies between 0 and $\frac{1}{2}$.
- For every N up to 2×10^5, $\ln N - \sum_{n=2}^{N} \frac{1}{n}$ lies in $[0, 1]$. At $N = 2 \times 10^5$ it is about 0.4228, close to $1 - \gamma \approx 0.4228$, where γ is Euler's constant.
- The bound is generous: $1 + 30 \ln 10 \approx 70.1$, and in fact $\sum_{n=1}^{10^{30}} \frac{1}{n} \approx 69.65$. The harmonic series diverges, but extremely slowly.

Common slips

- Using the wrong half of the double inequality: divergence needs the *lower* bound on the sum, and the final part needs the *upper* bound.
- Forgetting the $n = 1$ term: the inequalities are about $\sum_{n=2}^{N}$, while the conclusions are about $\sum_{n=1}^{N}$.
- Not justifying that the integrand's maximum is at $t = 1$. Saying it is increasing (or simply that $t \leq 1$ and $n - t \geq n - 1$) is all it takes.

Beyond STEP

The quantity $\sum_{n=1}^{N} \frac{1}{n} - \ln N$ decreases and is bounded below, so it converges. Its limit is Euler's constant $\gamma \approx 0.5772$, and nobody yet knows whether γ is rational. Comparing a sum with an integral, as we did here, is the *integral test*, one of the first tools for series in a first-year analysis course. Abbott's *Understanding Analysis* is a good place to see it made rigorous.

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